

Assignment

Tableau Visualizations

Tableau Practical File: - [Code](#)

Part 1: Individual Visualization Sheets (Tasks 1–7)

Task 1: Category Performance Analysis

- **Goal:** Compare total Sales by Category and identify the top performer.
- **Tableau Steps:**
 1. Drag Category to the **Rows** shelf.
 2. Drag Sales to the **Columns** shelf.
 3. Click the **Sort Descending** button on the toolbar so the highest-earning category sits at the top.
 4. Hold Ctrl and drag Sales onto the **Label** card to show the exact revenue values.
- **Recommended Chart:** Horizontal Bar Chart.

Task 2: Sales Trend Analysis

- **Goal:** Analyse monthly Sales trends to identify peak and slow periods.
- **Tableau Steps:**
 1. Drag Order Date to the **Columns** shelf. Click its dropdown menu and select **Month** (the second "Month" option in the list that shows *Month Year*, like "May 2015", to keep it continuous).
 2. Drag Sales to the **Rows** shelf.
 3. Change the mark type to **Line** in the Marks card if it hasn't automatically done so.
- **Recommended Chart:** Continuous Line Chart.

Task 3: Regional Contribution Analysis

- **Goal:** Visualize Region-wise Sales contribution to find the highest contributor.
- **Tableau Steps:**
 1. Change the Marks card dropdown from *Automatic* to **Pie**.
 2. Drag Region to the **Colour** card.
 3. Drag Sales to the **Angle** card.

4. Drag both Region and Sales to the **Label** card.
 5. *Optional Percentages:* Right-click the Sales pill on the Label card, select **Quick Table Calculation** $\rightarrow$ **Percent of Total**.
- **Recommended Chart:** Pie Chart or Donut Chart.

Task 4: Profitability Investigation

- **Goal:** analyse the relationship between Sales and Profit to spot loss-making orders.
- **Tableau Steps:**
 1. Drag Sales to the **Columns** shelf.
 2. Drag Profit to the **Rows** shelf.
 3. Go to the top menu: **Analysis** $\rightarrow$ uncheck **Aggregate Measures**. This plots every individual order record as a separate dot.
 4. Drag Profit onto the **Colour** card. Click *Edit Colours*, choose the *Red-Green Diverging* palette, and centre it at 0. Any loss-making orders will instantly pop out in bright red.
- **Recommended Chart:** Scatter Plot.

Task 5: Business Growth Evaluation

- **Goal:** Visualize cumulative Sales growth over time.
- **Tableau Steps:**
 1. Drag Order Date (Continuous Month or Day) to the **Columns** shelf.
 2. Drag Sales to the **Rows** shelf.
 3. Right-click the Sales pill on the Rows shelf, navigate to **Quick Table Calculation**, and select **Running Total**.
- **Recommended Chart:** Cumulative Line Chart / Running Total Chart.

Task 6: Category vs. Region Performance

- **Goal:** Compare Category-wise Sales across Regions to see strong/weak markets.
- **Tableau Steps:**
 1. Drag Category and Sub-Category to the **Rows** shelf.
 2. Drag Region to the **Columns** shelf.

3. Drag Sales to the **Color** card.
 4. Change the Mark type to **Square**. This creates a matrix where the color depth indicates the sales volume, making weak spots immediately obvious.
- **Recommended Chart:** Heatmap / Highlight Table.

Task 7: Product Prioritization

- **Goal:** Multi-dimensional analysis using Sales, Profit, and Quantity to identify priority items.
- **Tableau Steps:**
 1. Drag Sales to the **Columns** shelf and Profit to the **Rows** shelf.
 2. Uncheck **Analysis** $\rightarrow$ **Aggregate Measures** (or drag Product Name / Sub-Category to the **Detail** card).
 3. Drag Quantity onto the **Size** card. Larger bubbles now represent higher volume.
 4. Drag Profit onto the **Color** card (Red-Green Diverging) to easily distinguish highly profitable products from liabilities.
- **Recommended Chart:** Bubble / Enhanced Scatter Plot.

Part 2: Dashboard Creation (Task 8)

Task 8: Executive Dashboard Creation

1. Click the **New Dashboard** icon at the bottom of the workbook.
2. Set the Size to **Automatic** or **Desktop Browser** so it scales nicely for the CEO.
3. Drag at least three key sheets onto the dashboard workspace (e.g., *Task 1: Bar Chart*, *Task 2: Line Chart*, and *Task 4: Scatter Plot*).
4. Click on any of the sheets inside the dashboard view, click the small down arrow in its top-right options border, and select **Filters** $\rightarrow$ **Region** (or **Category**).
5. Click the down arrow on the newly appeared Filter card and select **Apply to Worksheets** $\rightarrow$ **All Using This Data Source**. Now, when the CEO filters by a region, the entire dashboard updates dynamically!

Part 3: Written Answers (Q9 & Q10)

Q9. Chart Selection Justification

Explain why different charts were used for different business questions in this analysis.

- **Bar Charts (Category Performance):** Best for discrete categorical comparisons. The human eye naturally evaluates differences in bar lengths instantly, making it perfect for spotting the top-earning category.
- **Line Charts (Sales Trends & Cumulative Growth):** Ideal for continuous data over time. Chronological connection points highlight upward/downward trajectories, seasonal spikes, and business velocity.
- **Pie Charts (Regional Contribution):** Excellent for showing "part-to-whole" relationships when categories are limited (e.g., 4–5 regions), allowing managers to see structural market share at a glance.
- **Scatter Plots (Profitability & Prioritization):** Essential for multi-dimensional relationship testing. By mapping independent variables against axes and injecting size and color, we compress 3 or 4 dimensions of data (Sales, Profit, Volume) into a single, highly readable view.

Q10. Business Insight Summary

Summarize two key insights from your dashboard and explain how they can help management make better decisions.

(Note: Because the specific values depend on your underlying data, adapt these structural templates based on what your charts reveal):

1. Insight 1: Volume vs. Profit Margin Divergence

- *Finding:* The Scatter Plot reveals that while certain Sub-Categories (e.g., "Tables" or "Supplies") generate high volume and high sales, they fall drastically below the \$0\$ profit line, generating heavy losses.
- *Actionable Decision:* Management can use this to adjust pricing strategies, renegotiate vendor costs, or scale back marketing promotions on those specific loss-leading items to protect overall margins.

2. Insight 2: Regional Seasonality Patterns

- *Finding:* Combining the regional filter with the monthly sales trend line reveals that the *West* region experiences massive spikes in Q4, while the *South* region remains completely stagnant during the same period.
- *Actionable Decision:* Supply Chain and Marketing can reallocate inventory and promotional budgets away from stagnant regions and funnel them directly into high-momentum regions ahead of peak quarters to maximize ROI.

